

39 Specific monoclonal antibodies can be used to treat some forms of cancer. One example is trastuzumab, which can be used in the treatment of tumours caused by cancer.

Which statements could help to explain why monoclonal antibodies such as trastuzumab are suitable for this role?

- 1 The variable regions of monoclonal antibodies such as trastuzumab can change, allowing the antibodies to bind to cancer cells even if their antigens mutate.
- 2 Monoclonal antibodies such as trastuzumab bind to specific cell surface receptors on cancer cells, which prevents these receptors from receiving signals that are needed for cancer cells to grow and divide.
- 3 The antigen-binding sites of monoclonal antibodies such as trastuzumab are complementary to antigens found on some cancer cells, and have stable tertiary structures that do **not** change.

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only